

Supplementary Material — Rank Before You Merge: A Stage Law for Training-Free Visual Token Compression in Merger-Equipped Vision-Language Models

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Every number is indexed by an audit identifier in S9. Tables S1–S3/S7 support § 5 of the body; S4–S6 support § 4; S8 supports § 3; S9 gives the per-cell artifact index, with cell, sample count, official metric, token count, skip accounting, and manifest identifier in S9.1–S9.3; S10 is a qualitative same-image pre/post selection comparison rendered from measured Qwen3-VL-8B merger features (four panels each, no paired question).

Model	Benchmark	Δ @75%	Δ @50%	Δ @25%
Qwen3-VL-8B	TextVQA	+8.7 pp	+29.0 pp	+38.3 pp
Qwen3-VL-8B	DocVQA	−1.3 pp	+5.9 pp	+26.5 pp
Qwen2.5-VL-7B	TextVQA	+7.0 pp	+15.3 pp	+32.0 pp
Qwen2.5-VL-7B	DocVQA	−2.1 pp	−1.4 pp	+18.8 pp

S1. M1 — MERGER RANKING RESHUFFLE, FULL TABLE (QWEN3-VL-8B, SEED-0, $n = 64$ /BENCHMARK)

Per image, Spearman correlation of the pre ranking (merger-input unit L2) with the post ranking (merged-token L2) over all units, plus top-25% kept-set overlap. Evidence record: audit index entry for the M1 mechanism campaign. The Kendall- τ column is taken from the same audited M1 record.

Benchmark	Spearman ρ	Jaccard@25%	Kendall τ
DocVQA (document)	0.137	0.180	0.094
TextVQA (scene)	0.332	0.243	0.227
GQA (object)	0.360	0.278	0.249

Reading: the reshuffle is largest on the most text-dense benchmark and smallest (but still substantial) on object-centric GQA — $\rho = 0.137 < 0.332 < 0.360$; Jaccard = $0.180 < 0.243 < 0.278$. For two independent size- κN subsets, the large- N chance reference is $\kappa/(2 - \kappa) = 1/7 \approx 0.143$ at $\kappa = 25\%$. All three overlaps are above chance but low in absolute terms, with DocVQA closest to chance. Per-image distributions are in Supplementary Figure 2 (top).

S2. M2 — DEMOTION DIRECTIONALITY VS TEXT-STROKE ENERGY, FULL TABLE (SAME SAMPLE AS S1)

rank_shift = post_rank − pre_rank (+ $\Rightarrow$ demoted); per-unit Sobel edge energy as the text-stroke proxy. Evidence record: anonymized artifact manifest entry for the M2 mechanism campaign. The per-benchmark Sobel means and median fractions are taken from the same audited M2 record.

Benchmark	ρ	Sobel drop	Sobel reverse	demoted > median
DocVQA	+0.439	0.641	0.124	92% (vs 35%)
TextVQA	+0.155	0.281	0.186	72% (vs 53%)
GQA	+0.036	0.304	0.271	58% (vs 53%)

Reading: the merger preferentially demotes high-edge (text-stroke) units $\sim 12\times$ more strongly on documents than on object scenes; on DocVQA the units post drops that pre keeps are the highest-edge of all (92% above the per-image median vs 35% for the reverse group) — post-merger selection is systematically anti-text on documents. Directionality hypothesis (merger trained on natural content attenuates high-frequency stroke energy): testable hypothesis, not a measured result. Second text-stroke proxy (Laplacian variance): future work (single-proxy disclosure).

S3. RETENTION CURVE — FULL NUMBER STRINGS ($n = 200$, BOTH QWEN MODELS)

pre-post Δ at 75% / 50% / 25% retention (the body § 5.4 prints endpoints only). Evidence record: audit index entry for the retention-curve ablation.

Reading: the two stages are indistinguishable under shallow compression (DocVQA post even leads 1–2 pp within noise at 75%); the pre-merger lead emerges and widens monotonically with depth. Same shallow regime as the Qwen2.5-VL 600k-cap reversal (body § 5.4 boundary; § 8 item 3). The Qwen3-VL curves are visualized in Supplementary Figure 2 (bottom).

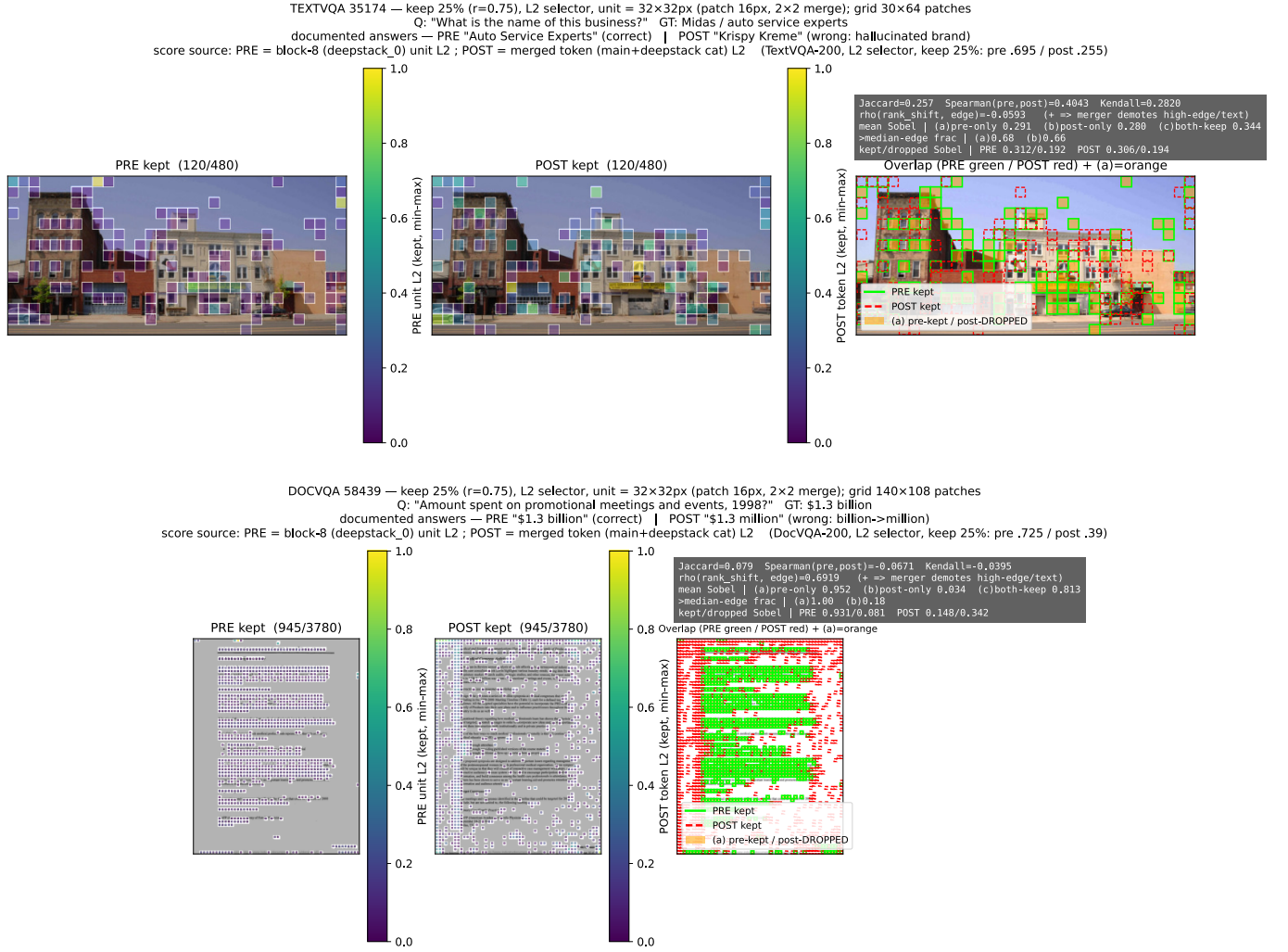


Fig. 1. **Representative token-survival maps at 25% retention.** TextVQA (top) and DocVQA (bottom) use the same query-blind L2 selector before and after Qwen3-VL's native merger. In each row, the first two panels show the pre- and post-selected units; the third overlays pre-kept (green), post-kept (red dashed), and pre-only/post-dropped units (orange). These are audited single-image mechanism examples, not aggregate accuracy evidence.

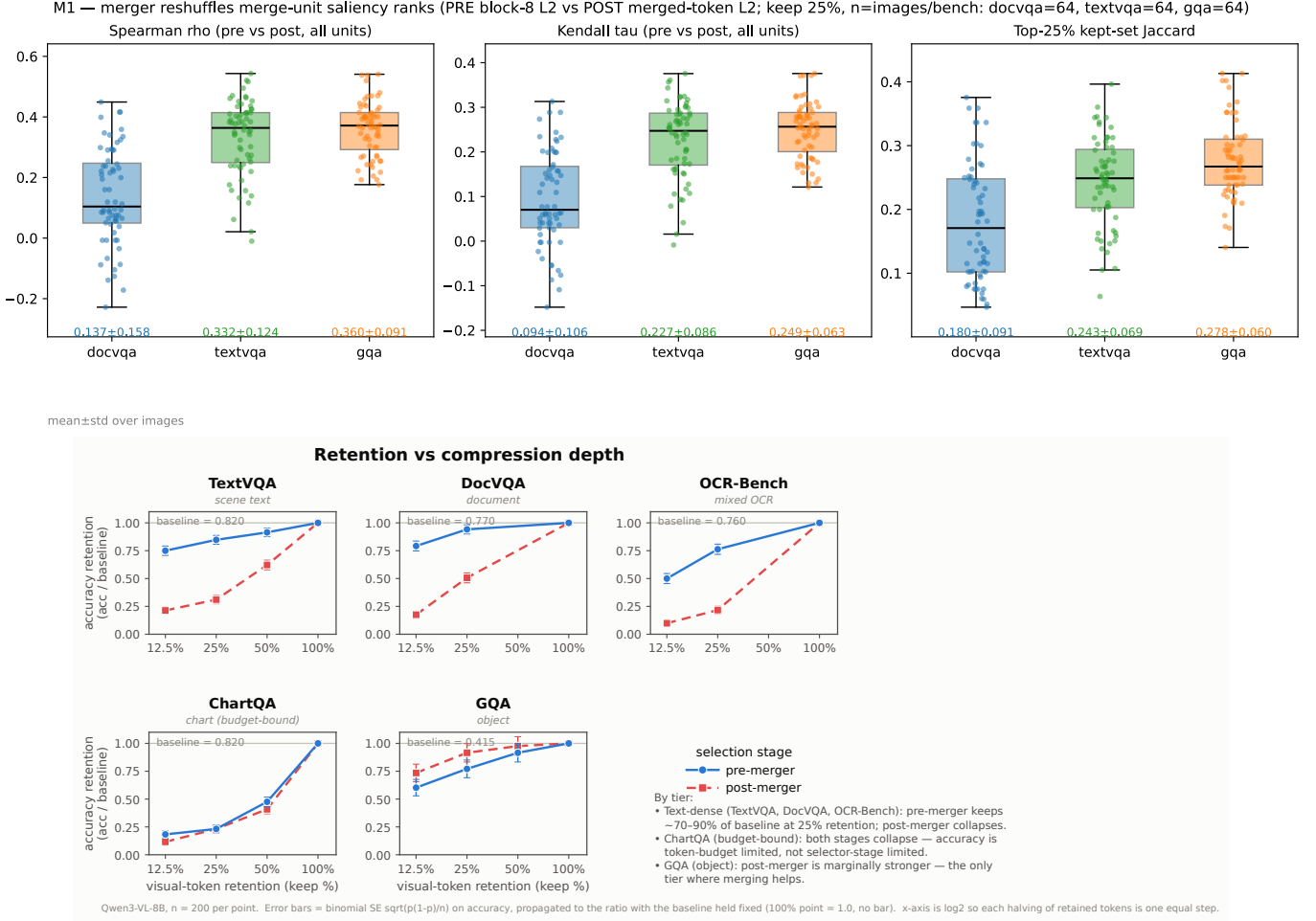


Fig. 2. **Quantitative mechanism and depth diagnostics.** Top: per-image pre/post rank agreement on the deterministic seed-0 M1 sample ($n = 64$ per benchmark); boxes and points show the distributions, with mean±standard deviation annotated. Bottom: normalized accuracy as compression deepens on Qwen3-VL-8B ($n = 200$ per point). Pre-merger selection preserves substantially more text-dense accuracy, while ChartQA is primarily budget-limited and GQA does not exhibit the text-dense stage effect. Error bars are descriptive rate-approximation standard errors; primary inference uses paired metric-level bootstrap and sign-flip tests.

S4. PAIRED McNEMAR CROSS-CHECKS — PER CELL (FULL SPLITS, BOTH QWEN MODELS)

Binary-outcome cross-check = paired McNemar z on per-sample correctness, $z = (b - c) / \sqrt{b + c}$, with b/c the pre-only / post-only correct counts; positive z = pre-merger lead, negative = post-merger lead (GQA only). Cross-check = independent-binomial z (SE $\sqrt{se_{pre}^2 + se_{post}^2}$); “agree %” = per-question agreement of the recorded correctness indicators. Evidence record: audit index entry for the paired-test table; paired by question id, restricted to ids answered in both arms; the 200/200 ground-truth self-test is retained in the audit record.

† the only significant post-stage lead. These GQA cells use the deepstack[0]-input (layer-8) pre ranking; a pure-stage control that ranks at the main-merger input (pre-final) gives 0.0 pp on GQA (body § 4.2, S7b), so the apparent post>pre is a layer-8 ranking artifact, not a true stage reversal. GQA official word-normalized exact-match rescore gives a concordant McNemar $z = -8.1 / -7.1$ (Qwen3-VL / Qwen2.5-VL). Smallest text-dense $\Delta_{25} |z| = 14.6$ (Qwen2.5-VL OCR-Bench); smallest text-dense $\Delta_{12.5} |z| = 15.0$ (Qwen2.5-VL OCR-Bench); smallest $|z|$ anywhere = 5.7 (Qwen2.5-VL GQA). On OCR-Bench @25% the independent-binomial cross-check (18.2 / 14.7) marginally exceeds McNemar (16.7 / 14.6) because the binary correctness indicator coarsens the /1000 item score (body footnote e); McNemar remains a cross-check and is $\geq$ -consistent on every other cell.

S4b. Selector-scorer and stage-router gates (Qwen3-VL-8B, official rescore)

S4b.1 Frequency-aware selector (body § 6d): the mechanism-side probe of body § 3, $s_{freq}(u) = \alpha z(\|f_u\|_2) + \beta z(\text{Var}_u)$, tuned in two grid rounds on TextVQA₅₀₀ at 25% retention ($\alpha=1, \beta=0.6$; official VQA-acc). The variance term helps only the most text-dense cell and only under aggressive compression; everywhere else it is neutral or negative, so no no-regression-across-four-benchmarks gate passes. Evidence record: anonymized audit index entry for the frequency-aware selector.

S4b.2 Adaptive stage router (body § 6e): query-blind per-image workload detector on merger-input features — high-frequency tail ratio $r_{HF} = \Pr[\text{Var}_u > \text{Var} + \sigma_{\text{Var}}]$ and 20-bin L2-entropy $H(\|f_u\|_2)$ — routing each image to PRE (RBM, rank-before-merge) or POST (post-merger L2), both iso-budget. κ is the kept fraction; the winner of a $\tau_{HF} \times \tau_{ent}$ grid on TextVQA₅₀₀ ($\tau_{HF}=0.08, \tau_{ent}=2.0$) routes every image to PRE, reproducing plain RBM bit-for-bit: text benchmarks show 0.00 pp regression but GQA gain is also +0.00 pp (below the required +1 pp). The $\tau_{HF}=0.13$ threshold does route some scene images to POST but collapses TextVQA (51/320 flipped, -7 pp) while flipping only 22/243 GQA images (0.428 vs RBM 0.438). GQA and TextVQA r_{HF} distributions overlap (means 0.127 vs 0.142), so no single global threshold reconciles the two acceptance criteria. Evidence record: audit index entry for the adaptive stage-router gate and workload-signal analysis.

S4b.3 Exploratory learned boundary scorer (body § 6): we test whether a lightweight learned selector can improve on plain L2 without moving selection past the merger. A small MLP receives only pre-merger boundary features available at inference time: unit L2 norm, Sobel edge energy, local feature variance, normalized position, and neighborhood statistics. Its teacher is the post-merger importance ordering captured on separate training images; evaluation uses disjoint held-out benchmark samples and the official task scorers. Both learned and L2 arms retain exactly 25% of merger-input units and otherwise share the same Qwen3-VL-8B inference configuration.

The asymmetric task result is diagnostic rather than a method improvement. M2 independently shows that the post-merger ordering demotes text-stroke-rich units; distillation transfers this preference into a pre-merger scorer and causes a 16.0-pp TextVQA loss. The positive GQA result shows that the learned head is capable of changing task behavior, ruling out a trivial “model did not learn” explanation. We therefore retain plain L2 and do not use the learned scorer in any headline table.

S5. DESCRIPTIVE RATE-SCALE ERRORS + SKIP ACCOUNTING (MAIN FULL-SPLIT QWEN BLOCKS)

For binary exact-match/containment outcomes, $\sigma = \sqrt{p(1-p)/n}$ is the Bernoulli standard error. For VQA accuracy and ANLS, which retain partial credit, applying the same expression to the printed mean is only a descriptive rate-scale approximation, not an inferential error bar. All claims use the paired metric-level bootstrap and sign-flip tests in the main paper. Evidence record: audit index entry for the Qwen full-split table and skip accounting.

- Qwen3-VL none DocVQA: re-run at max-model-len 49,152, native resolution, **0/5349 skips** (ANLS 0.956).

- OCR-Bench none cells skip **18/1000 (Qwen3-VL) / 24/1000 (Qwen2.5-VL)** long-OCR images on context overrun, scored 0 (conservative); compressed cells skip ≤ 5 . The uncompressed anchor is therefore understated and the relative gaps are conservative.
- Qwen2.5-VL OCR-Bench @25% RBM: 4 M-pixel cap (mean tokens 229.6 vs post 282.0) $\rightarrow$ its +293-pt gap is conservative (the one iso-token exception in the Qwen block of Table 1, note d).

Per-cell descriptive σ values (full splits, computed from the printed official metric p and full-split n):

These values provide scale and skip-accounting context only; they are not used to test partial-credit metric differences. The full-split GQA post lead is an order of magnitude smaller than the text-dense gaps and is reframed by the pre-final control (S7b) as a layer-8 ranking artifact rather than a true stage reversal. ^a Qwen3-VL none-DocVQA native-resolution anchor: full-split run partly skipped on context overrun (re-run at max-model-len 49,152 gives ANLS 0.956, 0/5349 skips); σ not printed for the partial native cell. ^c GQA evaluated at $\kappa = 0.25$ only on the full split (no full-split $\kappa = 0.125$ cells; body footnote c).

- Qwen3-VL OCR-Bench none = 760/1000 over 982 attempted (18 skips scored 0); compressed @25%: 547 (pre) / 184 (post), skip ≤ 5 — ratio 3.0 $\times$, widening to 6.6 $\times$ at 12.5% (350 vs 53).

S6. $n = 200$ CROSS-SPLIT CONSISTENCY (DEV-SUBSET VS FULL-SPLIT DIRECTION)

The $n = 200$ dev scope used by the regime-map and mechanism probes is direction-consistent with the full splits. Evidence record: audit index entry for the cross-split consistency campaign. $\Delta_{25} = \text{pre} - \text{post}$ at $\kappa = 0.25$.

Reading: all four text-dense cells agree in sign (pre-merger lead) across the two scopes; magnitudes are within run noise (the $n = 200$ subset SE is $\approx 8\times$ larger, so the subset is retained only as a direction check, never as the headline). The full Qwen3-VL $n = 200$ cross-method view (RBM vs FastV vs Pyramid) and the compact Qwen2.5-VL $n = 200$ cross-generation matrix appear in the main paper’s regime table and the archived audit index, respectively.

S7. SELECTOR INVARIANCE (SECOND SCORER FAMILY, QWEN3-VL)

The stage law under a second, non-L2 scorer (global-centroid attention). Evidence record: audit index entry for the selector-invariance campaign.

Reading: invariant in sign and magnitude on Qwen3-VL — the effect is not an L2 artifact. On Qwen2.5-VL the L2 sign is invariant but the attention proxy degrades (proxy-family specificity, not a counter-example); L2 remains the paper’s selector, frozen as plain RBM (§ 3.2).

S7b. Pure-stage confound control — pre-final (Qwen3-VL-8B, $n = 200$)

The body § 4.2 pre-final control isolates the *stage* from the feature-space confound of the original pre (deepstack[0]-input, layer 8) vs post (main-merger output) contrast. pre-final ranks L2 at the main-merger’s *own input* (final ViT-block features) and selects the top- κ merge-units from the visual output; the deepstack mergers run untouched. The only difference from post is then before vs after the lossy 2×2 main merger — a pure stage variable. Iso-token verified (prompt token ids match 200/200 per benchmark for pre-final vs post). Evidence record: anonymized audit index entry for the pre-final control and its per-sample audit.

Reading: on text-dense benchmarks pre-final still leads post by a large margin (the text-dense stage effect is a true stage effect, not a deepstack[0]-input vs main-merger-output feature-space artifact). On GQA pre-final equals post (0.0 pp): there is no stage effect on object-QA, and the original pre-vs-post apparent post>pre (−2.8 pp Qwen3-VL, −2.6 pp Qwen2.5-VL; S4) was an artifact of the deepstack[0]-input layer-8 ranking being poorly suited to object-QA, not a true stage reversal. Consistent with the mechanism: the merger does not preferentially damage object features (M2 $\rho = 0.04$ on GQA, S2). pre-final tested on Qwen3-VL only (the deepstack confound is Qwen3-specific; Qwen2.5-VL shares the architecture but was not re-tested).

S8. MERGER TAP POINTS AND M-ROPE CONFIGURATION DETAILS (SUPPORTS § 3)

- **Qwen3-VL-8B:** the first deepstack merger has index 0 and the visual indexes are [8, 16, 24], so unit scores use **ViT block-8 outputs** (intermediate depth); the single resulting mask is shared by all mergers,

Model	Benchmark	κ	Δ (official)	McNemar z (check)	b (pre-only)	c (post-only)	agree %	indep-binom z
Qwen3-VL-8B	TextVQA	.25	+38.4 pp	+43.0	2349	183	49.4	42.3
Qwen3-VL-8B	DocVQA	.25	+24.3 pp	+34.7	1601	150	67.3	27.1
Qwen3-VL-8B	OCR-Bench	.25	+363 pts	+16.7	422	56	52.2	18.2
Qwen3-VL-8B	GQA	.25	-2.8 pp †	-8.0	834	1196	83.9	-4.5
Qwen3-VL-8B	TextVQA	.125	+34.0 pp	+40.8	2101	161	54.8	39.9
Qwen3-VL-8B	DocVQA	.125	+24.9 pp	+33.1	1437	127	70.8	32.1
Qwen3-VL-8B	OCR-Bench	.125	+297 pts	+16.0	323	25	65.2	17.8
Qwen2.5-VL-7B	TextVQA	.25	+26.1 pp	+30.8	1682	309	60.2	27.3
Qwen2.5-VL-7B	DocVQA	.25	+11.0 pp	+15.6	1406	693	60.8	11.6
Qwen2.5-VL-7B	OCR-Bench	.25	+293 pts	+14.6	347	54	59.9	14.7
Qwen2.5-VL-7B	GQA	.25	-2.6 pp †	-5.7	892	1150	83.8	-4.1
Qwen2.5-VL-7B	TextVQA	.125	+27.8 pp	+33.1	1836	305	57.2	29.1
Qwen2.5-VL-7B	DocVQA	.125	+21.0 pp	+25.6	1325	294	69.7	23.3
Qwen2.5-VL-7B	OCR-Bench	.125	+268 pts	+15.0	294	26	68.0	15.9

bench@ κ	freq($\alpha=1, \beta=0.6$)	RBM (L2)	Post-L2	Δ freq-RBM
TextVQA @25%	0.6173	0.6053	0.2073	+1.20
TextVQA @50%	0.6780	0.6733	0.3873	+0.47
DocVQA @25%	0.4903	0.5063	0.2317	-1.60
DocVQA @50%	0.6271	0.6212	0.5487	+0.59
OCR-Bench @25%	0.7160	0.7380	0.1700	-2.20
OCR-Bench @50%	0.8220	0.8220	0.4980	0.00
GQA @25%	0.4220	0.4380	0.4540	-1.60
GQA @50%	0.4860	0.4840	0.5000	+0.20

TABLE I

FREQUENCY-AWARE SELECTOR (OFFICIAL METRICS, $n = 500$;
TEXTVQA=1.2 PP GAIN AT $\kappa=25\%$, BASELINE-EQUAL ELSEWHERE,
STRONG REGRESSION ON THE OTHER THREE BENCHMARKS AT 25%).

Benchmark	Learned POST teacher	Plain L2	Δ learned-L2
TextVQA	0.435	0.595	-16.0 pp
GQA	0.550	0.515	+3.5 pp

TABLE II

HELD-OUT LEARNED-SCORER EVALUATION AT 25% RETENTION ($n = 200$
PER BENCHMARK; OFFICIAL METRICS). THE SCORER CAN IMITATE A
POST-MERGER TEACHER, BUT THE RESULTING ORDERING IS STRONGLY
ANTI-TASK ON TEXTVQA AND THEREFORE FAILS THE
CROSS-WORKLOAD NO-REGRESSION GATE.

including the main merger that consumes the final-block output. The tap-point and call-order verification is retained in the audit index.

- **Qwen2.5-VL-7B**: no deepstack; scores are computed on the **main merger input (final ViT block output)**. `mask_computed_at='main'`.
- **InternVL3-8B**: the pixel-shuffle downsampling stage is the merger; L2 taps its input.
- **M-RoPE fix**: Qwen2.5-VL uses a block M-RoPE layout and Qwen3-VL an interleaved one; under pruning the position cursor must advance by the actual surviving count. A family-scoped fix (bit-degrading to the original at full retention) makes the Qwen2.5-VL compression path well-formed; the block-vs-interleaved contrast is itself a diagnostic of why naive pruning collapses one generation and is tolerated by the other.
- **r2c harness**: HF transformers eager attention, `--mropenative` (family-agnostic correct positions) on all same-scope cells; the Qwen2.5-VL $\times$ pre vllm-mimic positional degeneration (repeated “add-Criterion...” outputs $\rightarrow$ pseudo-zero scores) was detected and replaced before any reported number; on Qwen3-VL native vs mimic give 0.575 vs 0.525 (same ordering). The harness and position-layout verification are retained in the audit index.

S9. PER-CELL REPRODUCIBILITY INDEX

The filenames below define the anonymous artifact manifest and the audit records required for submission. The archive must be checked against this index before external upload; the tables remain self-contained summaries of the reported cells.

S9.1 InternVL3-8B full-split cells (16/16)

Model OpenGVLab/InternVL3-8B, vLLM 0.19, bf16, eager attention, greedy; $r = \text{TOTAL drop}$ (0.750 $\rightarrow \kappa = 25\%$, 0.875 $\rightarrow \kappa = 12.5\%$); skip = context-overrun cells scored 0. Evidence record: audit index entry for the InternVL3 full-split metrics and per-cell skip accounting. (Per-cell artifact identifiers are specified by the anonymous manifest. Pre-merger leads on every text-dense benchmark at both budgets — e.g. DocVQA ANLS 0.7284 vs 0.382 at $\kappa = 25\%$; GQA is the near-null object-centric case, 0.5993 vs 0.6031 — replicating the stage law on the pixel-shuffle merger design.)

S9.2 FastV-k3 same-scope cells (8/8)

FastV one-shot attention pruning at LLM layer $K = 3$, iso-budget $r = 0.75$ ($\kappa = 25\%$); HF-transformers 4.57.6 eager harness, official metrics. Official values are rescored outputs in the audit index; n , ptid, skip, engine, and per-cell identifiers are audited there. The harness’s raw `acc` field is not used as the official score. (Filenames marked `_n500` for the Qwen3-VL DocVQA / OCR-Bench cells ran at the $n = 200$ same-scope budget recorded in the JSON `n` field; OCR-Bench skips are long-OCR context overruns, scored 0.)

S9.3 RBM same-scope cells (5/5)

RBM (pre-merger L2) at iso-budget $r = 0.75$, HF-transformers eager, `--mropenative` (family-agnostic correct positions), matched to the FastV-k3 cells above. Official values, ptid, skip, and per-cell identifiers are recorded in the audit index.

S10. QUALITATIVE SAME-IMAGE COMPARISON — MEASURED PRE- VS POST-MERGER L2 SELECTION

For four representative inputs we render, on the *same* processor-resized image, unit grid, 25% budget and final visual-token count, the original image with its full merge-unit grid, the RBM (pre-merger L2 top- k) kept mask, the post-merger L2 top- k kept mask (same L2 selector, same budget), and a four-state difference map (kept-by-both / RBM-only / post-only / dropped-by-both, encoded by colour *and* hatch/outline so the states remain distinguishable in greyscale). Each panel reports the measured top- k Jaccard index $J = |\text{pre} \cap \text{post}| / |\text{pre} \cup \text{post}|$ and the full-ranking Spearman ρ between the pre- and post-merger L2 orderings (computed from the stored NPZ rank arrays; values cross-checked against the validation report, $|\Delta| < 10^{-5}$). All scores are the target model’s own merger-input / merger-output feature L2 norms (Qwen3-VL-8B-Instruct, `enforce_eager, max_pixels=1.5M`); no CLIP, SigLIP, attention map, or fabricated saliency is used. The FastV layer-2 panel is intentionally absent (no user-supplied question exists for these images; a query-conditioned mask cannot be measured without one) and is recorded as `fastv_status: skipped_missing_question`.

The consistently low Jaccard (0.25–0.35) and moderate Spearman (0.30–0.48) across the four shown inputs are the image-level signature of the body paper’s mechanism result: the learned 2×2 merger *reshuffles* the unit ranking, so selecting before versus after it on the same image with the same selector and budget keeps largely *different* units. The four-state maps show that the disagreement is spatially structured (contiguous RBM-only and post-only regions), not uniform noise, and that the “kept-by-both” intersection is a sparse minority of the budget.

Model	Benchmark	n	none	pre@25%	post@25%	pre@12.5%	post@12.5%
Qwen3-VL-8B	TextVQA	5000	0.0051	0.0069	0.0059	0.0071	0.0048
Qwen3-VL-8B	DocVQA	5349	— ^a	0.0068	0.0058	0.0065	0.0042
Qwen3-VL-8B	OCR-Bench	1000	0.0135	0.0157	0.0123	0.0151	0.0071
Qwen3-VL-8B	GQA	12578	0.0043	0.0044	0.0045	— ^c	— ^c
Qwen2.5-VL-7B	TextVQA	5000	0.0049	0.0065	0.0070	0.0069	0.0066
Qwen2.5-VL-7B	DocVQA	5349	0.0030	0.0066	0.0068	0.0068	0.0059
Qwen2.5-VL-7B	OCR-Bench	1000	0.0122	0.0158	0.0122	0.0149	0.0079
Qwen2.5-VL-7B	GQA	12578	0.0044	0.0044	0.0044	— ^c	— ^c

Model	Bench	full Δ_{25}	$n = 200$ Δ_{25}	sign
Qwen3-VL	TextVQA	+38.4 pp	+38.3 pp	✓ pre
Qwen3-VL	DocVQA	+24.3 pp	+26.5 pp	✓ pre
Qwen2.5-VL	TextVQA	+26.1 pp	+32.0 pp	✓ pre
Qwen2.5-VL	DocVQA	+11.0 pp	+18.8 pp	✓ pre

Selector	Benchmark	pre	post	Δ
L2	TextVQA	0.598–0.605	0.215–0.222	$\approx +38$ pp
GC-attn	TextVQA	0.553	0.200	+35.3 pp
GC-attn	TextVQA/DocVQA (2n)	—	—	+35.5 / +24.4 pp

Packaging note: this source package contains the rendered summaries above, not the run-level anonymous artifact indexed in S9. That archive must be restored, sanitized, checksummed, and verified before submission. This supervisor-review draft identifies Zhengxing Yan but includes no affiliation or repository identifier; a future anonymous submission requires a separate anonymization pass.

Benchmark (metric)	none	pre-final	post	Δ (paired mean $\pm$ SE)	verdict
TextVQA (VQA-acc)	0.858	0.515	0.215	+30.0 pp (+0.300 $\pm$ 0.039)	stage effect
DocVQA (ANLS)	0.976	0.326	0.200	+12.5 pp (+0.125 $\pm$ 0.032)	stage effect
GQA (exact match)	0.605	0.450	0.450	0.0 pp (+0.000 $\pm$ 0.032)	no stage effect

Body location	Cells	Artifact record	Audit summary
Main full-split Qwen blocks	26/26	audit: main-qwen	full-split metrics and skip accounting
Main full-split InternVL3 block	16/16	audit: main-internvl3	full-split metrics and skip accounting
Regime-map FastV-k3	8	audit: regime-fastv-k3	same-scope rescored cells
Regime-map RBM same-scope	5	audit: regime-rbm	same-scope rescored cells
Efficiency study	7 configs	audit: efficiency	throughput and latency records
§ 5 kept-set Jaccard / swap	8	audit: mechanism-swap	kept-set and ordering audit
§ 5.1–5.2 M1/M2	$n = 64 \times 3$	audit: mechanism-m1-m2	rank and edge-energy measurements
§ 5.4 retention curve	$n = 200 \times 4$	audit: retention-curve	depth ablation records
§ 6 (a)–(e) prespecified extensions	—	audit: negative-results	gates and extension records
Learned-scorer probe	$n = 200 \times 2$	audit: learned-scorer	model, held-out splits, predictions, and official rescoring
§ 8 600k reversal	3	audit: low-cap-reversal	configuration-boundary records
Engine fairness	$r=0$ 8/8; none 16/16	audit: engine-parity	zero-drop and none-cell checks
PyramidDrop iso-25% collapse	$n = 500$	audit: pyramiddrop-iso25	iso-budget baseline record

cell	mode	r	bench	n	official	metric	ptid	skip	JSON
none_docvqa	none	0.000	docvqa	5349	0.9221	ANLS	3230.8	0	internvl3_none_docvqa_r0.000_full.json
none_gqa	none	0.000	gqa	12578	0.6293	acc-official	2557.2	0	internvl3_none_gqa_r0.000_full.json
none_ocrbench	none	0.000	ocrbench	1000	0.852	Final/1000=852	2040.1	0	internvl3_none_ocrbench_r0.000_full.json
none_textvqa	none	0.000	textvqa	5000	0.8338	VQA-acc	2562.9	0	internvl3_none_textvqa_r0.000_full.json
pre_docvqa_r0.750	pre	0.750	docvqa	5349	0.7284	ANLS	859.5	0	internvl3_pre_docvqa_r0.750_full.json
pre_docvqa_r0.875	pre	0.875	docvqa	5349	0.5054	ANLS	464.3	0	internvl3_pre_docvqa_r0.875_full.json
pre_gqa_r0.750	pre	0.750	gqa	12578	0.5993	acc-official	690.1	0	internvl3_pre_gqa_r0.750_full.json
pre_ocrbench_r0.750	pre	0.750	ocrbench	1000	0.753	Final/1000=753	555.4	0	internvl3_pre_ocrbench_r0.750_full.json
pre_textvqa_r0.750	pre	0.750	textvqa	5000	0.789	VQA-acc	690.5	0	internvl3_pre_textvqa_r0.750_full.json
pre_textvqa_r0.875	pre	0.875	textvqa	5000	0.723	VQA-acc	378.4	0	internvl3_pre_textvqa_r0.875_full.json
post_docvqa_r0.750	post	0.750	docvqa	5349	0.382	ANLS	859.5	0	internvl3_post_docvqa_r0.750_full.json
post_docvqa_r0.875	post	0.875	docvqa	5349	0.2451	ANLS	464.3	0	internvl3_post_docvqa_r0.875_full.json
post_gqa_r0.750	post	0.750	gqa	12578	0.6031	acc-official	690.1	0	internvl3_post_gqa_r0.750_full.json
post_ocrbench_r0.750	post	0.750	ocrbench	1000	0.321	Final/1000=321	555.4	0	internvl3_post_ocrbench_r0.750_full.json
post_textvqa_r0.750	post	0.750	textvqa	5000	0.4148	VQA-acc	690.5	0	internvl3_post_textvqa_r0.750_full.json
post_textvqa_r0.875	post	0.875	textvqa	5000	0.3064	VQA-acc	378.4	0	internvl3_post_textvqa_r0.875_full.json

cell	model	bench	n	official	ptid	skip	engine	JSON
r2b_qwen3vl_fastv_k3_textvqa_r0.75_full5000	Qwen3-VL	textvqa	5000	0.7771	213.7	5	hf eager	r2b_qwen3vl_fastv_k3_textvqa_r0.75_full5000.json
r2b_qwen3vl_fastv_k3_gqa_r0.75_full12578	Qwen3-VL	gqa	12578	0.5376	96.8	0	hf eager	r2b_qwen3vl_fastv_k3_gqa_r0.75_full12578.json
r2b_qwen3vl_fastv_k3_docvqa_r0.75_n500	Qwen3-VL	docvqa	200	0.5863	176.0	0	hf eager	r2b_qwen3vl_fastv_k3_docvqa_r0.75_n500.json
r2b_qwen3vl_fastv_k3_ocrbench_r0.75_n500	Qwen3-VL	ocrbench	200	0.415	114.8	19	hf eager	r2b_qwen3vl_fastv_k3_ocrbench_r0.75_n500.json
r2b_qwen2vl_fastv_k3_textvqa_r0.75_n200	Qwen2.5-VL	textvqa	200	0.7467	282.1	0	hf eager	r2b_qwen2vl_fastv_k3_textvqa_r0.75_n200.json
r2b_qwen2vl_fastv_k3_gqa_r0.75_n200	Qwen2.5-VL	gqa	200	0.475	126.8	0	hf eager	r2b_qwen2vl_fastv_k3_gqa_r0.75_n200.json
r2b_qwen2vl_fastv_k3_docvqa_r0.75_n200	Qwen2.5-VL	docvqa	200	0.4852	231.4	0	hf eager	r2b_qwen2vl_fastv_k3_docvqa_r0.75_n200.json
r2b_qwen2vl_fastv_k3_ocrbench_r0.75_n200	Qwen2.5-VL	ocrbench	200	0.285	143.6	20	hf eager	r2b_qwen2vl_fastv_k3_ocrbench_r0.75_n200.json

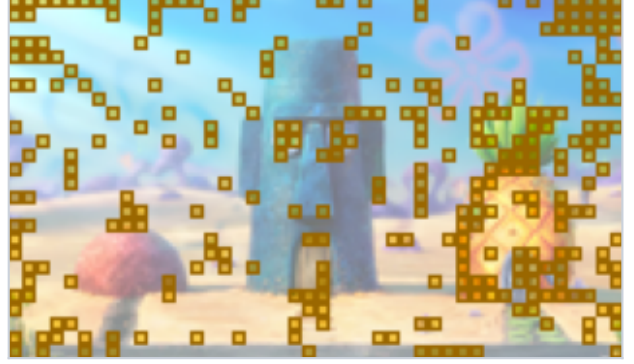
cell	model	bench	n	official	ptid	skip	engine	JSON
r2c_qwen2vl_pre_r0.75_textvqa_n200	Qwen2.5-VL	textvqa	200	0.6683	282.1	0	hf eager	r2c_qwen2vl_pre_r0.75_textvqa_n200.json
r2c_qwen2vl_pre_r0.75_gqa_n200	Qwen2.5-VL	gqa	200	0.520	126.8	0	hf eager	r2c_qwen2vl_pre_r0.75_gqa_n200.json
r2c_qwen2vl_pre_r0.75_docvqa_n200	Qwen2.5-VL	docvqa	200	0.5062	231.4	0	hf eager	r2c_qwen2vl_pre_r0.75_docvqa_n200.json
r2c_qwen2vl_pre_r0.75_ocrbench_n200	Qwen2.5-VL	ocrbench	200	0.370	143.6	20	hf eager	r2c_qwen2vl_pre_r0.75_ocrbench_n200.json
r2c_qwen3vl_pre_r0.75_ocrbench_n200	Qwen3-VL	ocrbench	200	0.575	114.8	19	hf eager	r2c_qwen3vl_pre_r0.75_ocrbench_n200.json

Same-image iso-budget comparison - df7282e1 (Qwen3-VL-8B measured merger L2)

a original



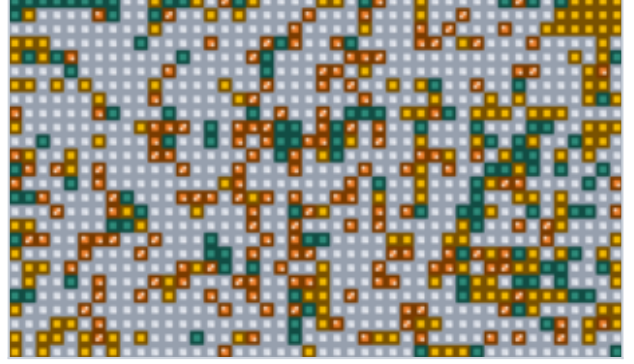
b RBM (pre L2)



c post L2



d pre vs post



N=1144 k=286 (25% kept) top-k Jaccard=0.2599 Spearman $\rho(\text{pre}, \text{post ranks})=0.3141$ unit grid $26 \times 44 = \text{grid_thw}/2$ Qwen3-VL-8B measured merger L2
FastV: SKIPPED (no user question; inputs/manifest.json absent)

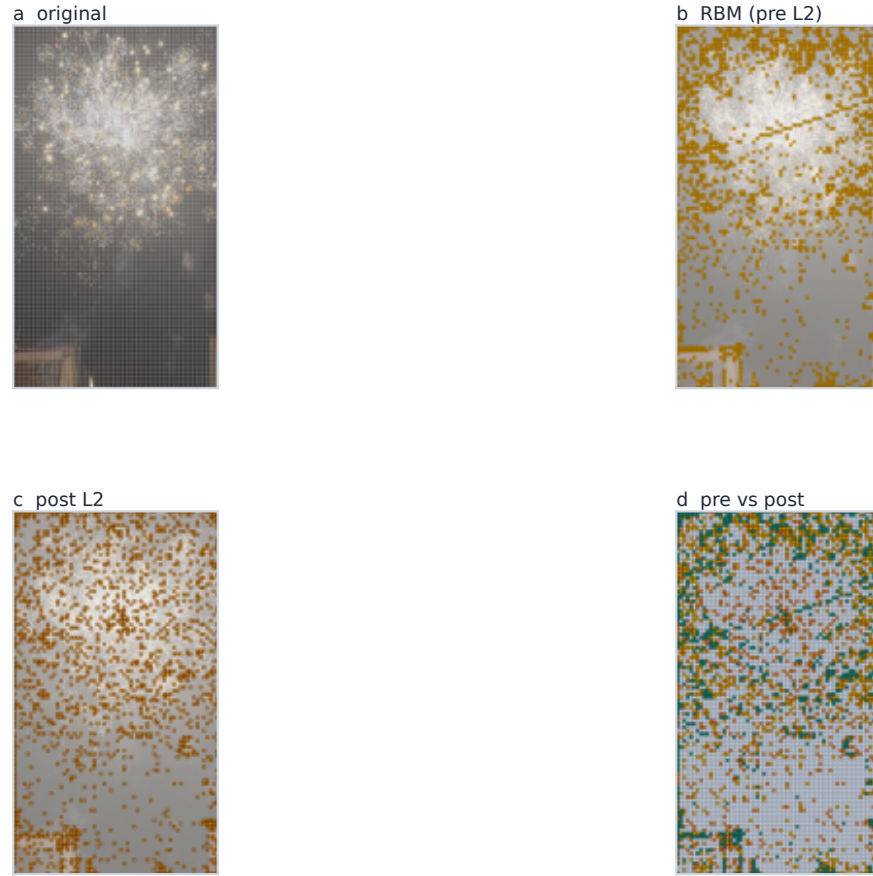
(a) full unit grid outline (light). (b,c) SAME amber L2 selector & 25% budget; (c) hatched = post (distinguishable without color).

(d) four states kept-by-both / RBM-only / post-only / dropped-by-both, encoded by color + hatch/outline (legend).

■ kept by both (pre & post) ■ RBM-only (pre & ~post) ■ post-only (~pre & post) ■ dropped by both

Fig. 3. Sample df7282e1 ($N=1144$, $k=286$, $J=0.260$, $\rho=0.314$). Sparse scene; the four-state map shows RBM-only (amber) concentrated on object boundaries while post-only (hatched) scatters over flat regions the merger smoothed.

Same-image iso-budget comparison - 69 (Qwen3-VL-8B measured merger L2)



$N=6820$ $k=1705$ (25% kept) top-k Jaccard=0.3478 Spearman $\rho(\text{pre}, \text{post ranks})=0.4774$ unit grid $110 \times 62 = \text{grid_thw}/2$ Qwen3-VL-8B measured merger L2
FastV: SKIPPED (no user question; inputs/manifest.json absent)

(a) full unit grid outline (light). (b,c) SAME amber L2 selector & 25% budget; (c) hatched = post (distinguishable without color).
(d) four states kept-by-both / RBM-only / post-only / dropped-by-both, encoded by color + hatch/outline (legend).

■ kept by both (pre & post) ■ RBM-only (pre & ~post) ■ post-only (~pre & post) ■ dropped by both

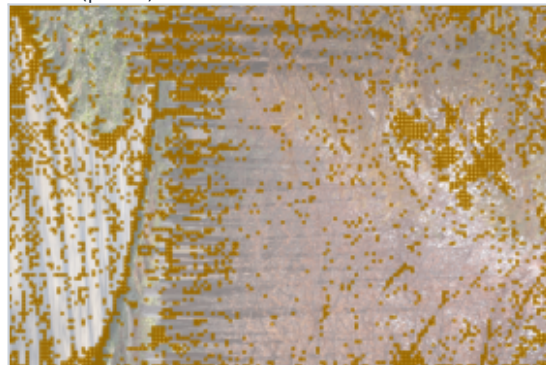
Fig. 4. Sample `img69` ($N=6820$, $k=1705$, $J=0.348$, $\rho=0.477$). Highest overlap of the four; the merger preserves more of the pre-ranking here, yet two-thirds of the kept set still differs.

Same-image iso-budget comparison - 72 (Qwen3-VL-8B measured merger L2)

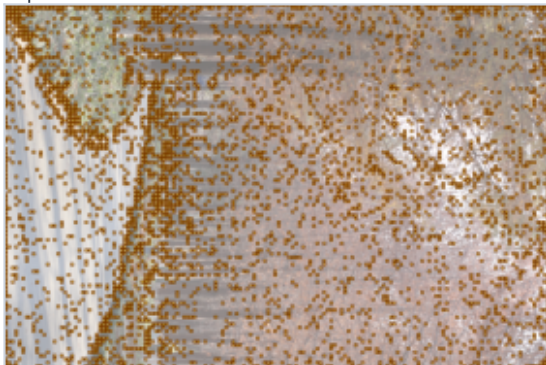
a original



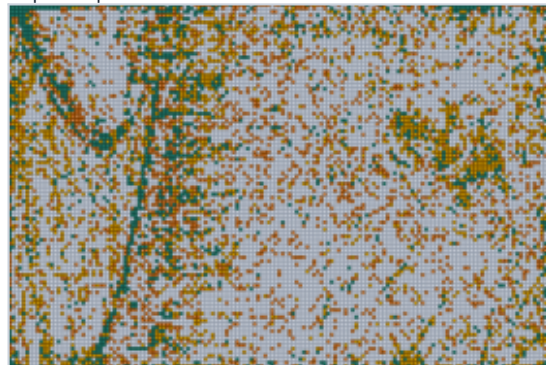
b RBM (pre L2)



c post L2



d pre vs post



$N=16224$ $k=4056$ (25% kept) top-k Jaccard=0.2482 Spearman $\rho(\text{pre}, \text{post ranks})=0.2999$ unit grid $104 \times 156 = \text{grid_thw}/2$ Qwen3-VL-8B measured merger L2
FastV: SKIPPED (no user question; inputs/manifest.json absent)

(a) full unit grid outline (light). (b,c) SAME amber L2 selector & 25% budget; (c) hatched = post (distinguishable without color).
(d) four states kept-by-both / RBM-only / post-only / dropped-by-both, encoded by color + hatch/outline (legend).

■ kept by both (pre & post) ■ RBM-only (pre & ~post) ■ post-only (~pre & post) ■ dropped by both

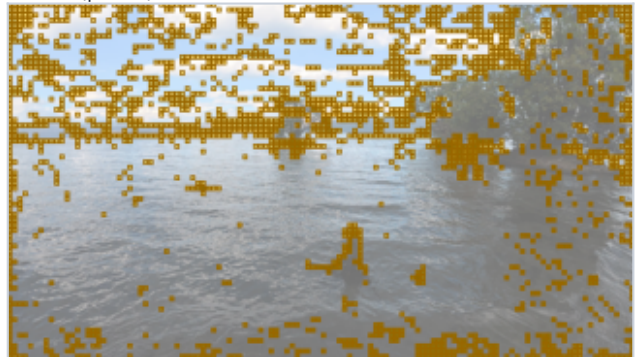
Fig. 5. Sample `img72` ($N=16224$, $k=4056$, $J=0.248$, $\rho=0.300$). Textured outdoor scene with a wall edge; the lowest overlap — the merger's lossy aggregation most strongly reorders the high-frequency edge vs. texture units.

Same-image iso-budget comparison - 74 (Qwen3-VL-8B measured merger L2)

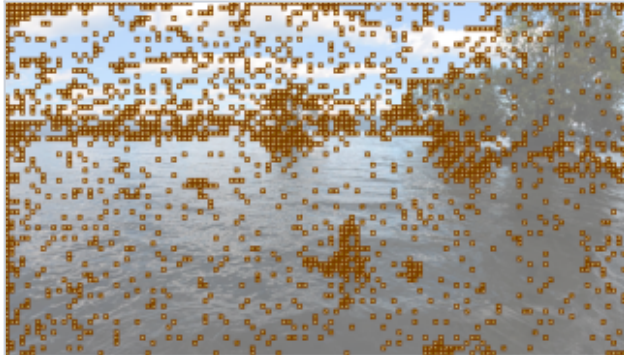
a original



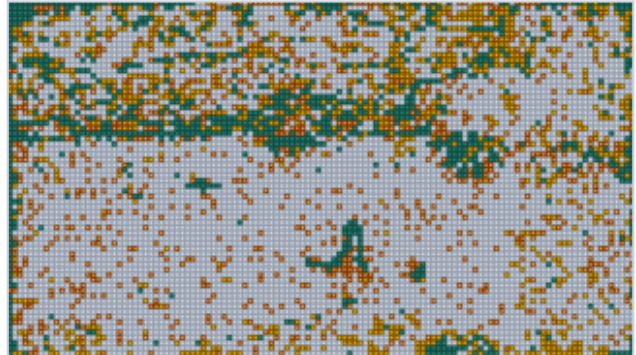
b RBM (pre L2)



c post L2



d pre vs post



$N=8160$ $k=2040$ (25% kept) top-k Jaccard=0.3377 Spearman $\rho(\text{pre}, \text{post ranks})=0.3963$ unit grid $68 \times 120 = \text{grid_thw}/2$ Qwen3-VL-8B measured merger L2
FastV: SKIPPED (no user question; inputs/manifest.json absent)

(a) full unit grid outline (light). (b,c) SAME amber L2 selector & 25% budget; (c) hatched = post (distinguishable without color).

(d) four states kept-by-both / RBM-only / post-only / dropped-by-both, encoded by color + hatch/outline (legend).

■ kept by both (pre & post) ■ RBM-only (pre & ~post) ■ post-only (~pre & post) ■ dropped by both

Fig. 6. Sample `img74` ($N = 8160$, $k = 2040$, $J = 0.338$, $\rho = 0.396$). Waterscape with horizon; RBM-only units track the water surface detail that the post-merger L2 demotes in favour of merged sky blocks.